

Determining critical phase transition temperature in an Ising model simulation

Bela F. Brunner¹ and Adam Zich¹

¹*Department of Physics,
Utrecht University*

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Room for abstract.

I. INTRODUCTION

A. The Ising model

The Ising model named after the German physicist Erns Ising is a model in statistical mechanics. The model consists of discrete magnetic spin dipole moments that take values ± 1 ("up" and "down" respectively) arranged on a lattice. For the purposes of our simulation we used a square lattice. The spins interact with neighbouring spins. The model exhibits phase transitions that can be associated with the gas-liquid transition or spontaneous magnetisation.

B. Aim of study

In this article we look at determining at what temperature does the Ising model undergo a phase transition. It is known from theory that under a certain critical temperature the model stabilises into completely up or down state. Conversely, above this critical, on average, the model remains a random mixture of up and down spins.

II. THEORY

The theoretical value of the critical temperature was predicted by Lars Onsager[1] to be

$$T_c = \frac{2}{\ln(1 + \sqrt{2})}. \quad (1)$$

The interaction energy between two spins is given by

$$E_{\text{int}} = -Js_i s_j, \quad (2)$$

where $J > 0$ is the coupling constant for ferromagnetic interactions and the s_i being the value of spin of spin i . We define the reduced interaction energy as $E_r = \frac{E_{\text{int}}}{J} = -s_i s_j$. The reduced interaction energy with the j nearest neighbours is then

$$E_r = \sum_{\langle j \rangle} s_i s_j, \quad (3)$$

III. METHODOLOGY

A. Metropolis algorithm

To simulate the evolution of the Ising model we use a Monte Carlo simulation using the Metropolis algorithm. In our implementation we consider the von Neumann neighbourhood of only 4 neighbouring spins in the $\pm x$ and $\pm y$ directions. We also use a periodic boundary condition such that if we denote by $n \times n$ the size of the lattice, we have that the spin at position (n, i) is a neighbor of spin at position $(1, i)$ and in the same way also spin at i, n is neighbouring the spin at $(i, 1)$. At each iteration of the algorithm a random position is chosen in the square lattice of spins. The energy needed to flip the spin from the up to the down state or vice versa is then calculated from Eq. (3) as

$$\Delta E = E_{\text{before}} - E_{\text{after}} = \sum_{\langle j \rangle} (-s_i) s_j - \sum_{\langle j \rangle} s_i s_j = -2 \sum_{\langle j \rangle} s_i s_j. \quad (4)$$

We then have 2 cases. If $\Delta E \leq 0$, we flip the spin orientation. If $\Delta E \geq 0$, we accept the flip with probability $p = e^{\frac{\Delta E}{T_r}}$. In order to do this, we sample a random variable $X \sim \text{Unif}(0, 1)$ and check $X < p$, in which case we accept the flip and leave the spin unflipped otherwise.

We know that for values of the reduced temperature below the critical temperature i.e. $T_r < T_c$ the system gets to an equilibrium state where the absolute value of the average magnetisation goes to 1[2]. This however takes a certain amount of spin-flip attempts before the equilibrium is reached. To calculate the average quantities $|\langle m \rangle|$ and C we then wait for a certain amount of spin-flip attempts before gathering values then used to calculate these quantities to give the model enough time to reach equilibrium.

B. Determining parameters for final simulation

For the simulation that produced Figure 1 we had a number of requirements. The first requirement was for the simulation that produces the figure we use in this article to take no longer than 20 minutes to run. This meant that we had to look for simulation parameters that produce good data but for the simulation runtime to stay within the time bound.

We already knew from test simulations that the critical temperature was somewhere in the range $1 < T_c < 3$ as

we ran simulation for $T_r = 1.5$ and $T_r = 3$ and observed the behaviour associated with reduced temperatures being lower and higher than the critical temperature respectively. This then set the range of reduced temperatures that we were going to simulate.

We then continued to choosing a size of the lattice that we want to simulate as this, we found, had impact on the number of spin-flip attempts needed for the model to reach equilibrium. We decided to use $n = 50$ as for values of $n < 20$ we found that the equilibrium was disturbed to a great degree by the non-zero value of the temperature causing fluctuations in the spins.

We then ran multiple simulations trying to ascertain the number of spin-flip attempts needed for the 50×50 lattice to reach equilibrium for the temperature range we wanted to study.

IV. RESULTS

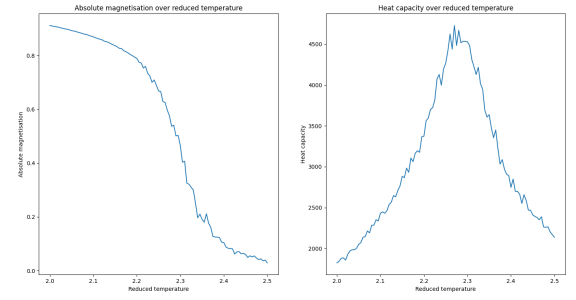


FIG. 1. Results

TODO: <Give results>
TODO: <Give figures of results>

V. COMPARISON TO THEORY

A. Theoretical solution for the critical phase transition temperature

TODO: <Give derivation of critical temperature in the Ising model>

B. Comparison of theory and our results

TODO: <Compare our results with theoreticla values>

VI. CONCLUSIONS

Conclusion.

[1] Lars Onsager. Crystal statistics. i. a two-dimensional model with an order-disorder transition. *Physical Review*, 65(3-4):117–149, 1944.

[2] Stephen J. Blundell and Katherine M. Blundell. *Concepts in Thermal Physics*. Oxford University Press, Oxford, 2nd edition, 2010.